



UTM

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SECI 1143 PROBABILITY AND STATISTICAL DATA ANALYSIS

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Introduction

Our analysis examines Malaysia's economic environment in 2022 by examining statistics on revenue and expenses from different states. We separated the states into Peninsular Malaysia and East Malaysia in order to comprehend regional differences. The dataset includes the Gini coefficient, average spending, average and median income, and state-specific poverty rates, among other important economic variables.

Since it represents the average income of all residents in each state, the **mean income** offers a broad picture of the state of the economy. Compared to the mean, the **median income** is less impacted by outliers and provides information about average income levels as it represents the midpoint of the income distribution. The average spending pattern and cost of living in each state are shown by **mean expenditure**. With values ranging from 0 (perfect equality) to 1 (perfect inequality), the **Gini coefficient** quantifies inequality in income. The percentage of people living below the poverty line is represented by the **poverty rate**, which reflects the level of poverty in each state.

In order to guarantee the accuracy of our analysis, we carried out multiple data prior treatment procedures. These include classifying states into East and Peninsular Malaysia for comparison analysis, handling null or missing results, and dealing with outliers. By choosing these factors, we make it possible to analyze the economic circumstances in each state and draw detailed comparisons between Peninsular Malaysia and East Malaysia. Our analysis uses regression, ANOVA, correlation, and hypothesis testing to try to give meaningful insights for improved decision-making.

Dataset

The dataset our group chose focuses on income and expenditure statistics for various states in Malaysia, categorized into East and Peninsular Malaysia for the year 2022. This dataset includes key variables such as the mean and median income, mean expenditure, Gini coefficient, and poverty rate for each state.

To prepare the data for analysis, we performed several preprocessing steps. First, we checked for and handled any missing or null values, ensuring the dataset was complete. Next, we identified and managed any outliers that might skew our analysis. Lastly, we categorized the states into East and Peninsular Malaysia to facilitate a comparative analysis.

We chose these variables because they provide a comprehensive view of the economic status across different states, allowing us to perform a detailed comparative analysis between East and Peninsular Malaysia. The **income mean** helps us understand the central tendencies and variations in income distribution. The **expenditure offers** insights into spending patterns and living costs.

By analyzing these variables, we aim to identify differences in economic well-being, and spending behavior between East and Peninsular Malaysia. Additionally, hypothesis testing, correlation, regression, and ANOVA are used to test the sample data. This analysis will provide valuable insights contributing to more informed decision-making.

3.0 Data Analysis

3.1 Hypothesis testing 2-sample

In this analysis, we use variable income mean of two samples which are Peninsular Malaysia and East Malaysia. This test is conducted to determine whether the mean of Peninsular Malaysia is different from the mean of East Malaysia or not at 95% confidence level, assuming unequal variances. From the data, frequency (n), mean ($\bar{x}$), standard deviation (s) are calculated.

From the calculation, we know that:

	Peninsular Malaysia	East Malaysia
Frequency	$n_1 = 13$	$n_2 = 3$
Mean	$\bar{x}_1 = 8120.23$	$\bar{x}_2 = 6959.33$
Standard Deviation	$s_1 = 3020.16$	$s_2 = 1126.86$

1. Hypothesis statement:

$$H_0 : \mu_1 = \mu_2$$

$$H_1 : \mu_1 \neq \mu_2$$

Where μ_1 equals the mean of Peninsular Malaysia and μ_2 equals the mean of East Malaysia.

2. Given 95% confidence level, $\alpha = 0.05$. The test statistics, t_0 can be calculated using the formula:

$$t_0 = \frac{\bar{x}_1 - \bar{x}_2 - 0}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

By using RStudio, the test statistic, $t_0 = 1.0945$

t_stat	1.09454850083883
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3. Calculate the degree of freedom using the formula:

$$v = \frac{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2} \right)^2}{\frac{\left(\frac{s_1^2}{n_1} \right)^2}{n_1 - 1} + \frac{\left(\frac{s_2^2}{n_2} \right)^2}{n_2 - 1}}$$

By using RStudio, degree of freedom, $v = 9.68 \approx 9$

v	9.68904379795111
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Therefore, using $\alpha = 0.05$, we reject the null hypothesis if $t_0 < t_{0.05,9}$ or $t_0 > t_{0.05,9}$.

From $\alpha = 0.05$, we find the p-value by using RStudio. We find that the p-value = 0.302

p_value	0.302141027179508
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$\therefore$ Critical value, $t_{0.05,9} = -2.2622$, $t_{0.05,9} = 2.2622$

t.alpha	-2.2621571627982
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4. Conclusion:

Since $t_0 = 1.0945 < t_{0.05,9} = 2.2622$, we fail to reject the null hypothesis. The p-value also shows that we fail to reject the null hypothesis since the p-value = 0.302 $> \alpha = 0.05$. Hence, there is no sufficient evidence to reject the claim that the mean of Peninsular Malaysia is different to the mean of East Malaysia.

3.2 Correlation Test

In this analysis, we will use variable income mean and expenditure mean of each state to determine whether there is a linear relationship between both of the variables. Since both of the variables are ratio type, we choose to apply Pearson's Product-Moment Correlation to calculate the sample correlation coefficient, r .

The formula for calculating the sample correlation coefficient is as follows;

$$r = \frac{\sum xy - (\sum x \sum y) / n}{\sqrt{[(\sum x^2) - (\sum x)^2 / n][(\sum y^2) - (\sum y)^2 / n]}}$$

where:

r = Sample correlation coefficient

n = Sample size

x = Value of the independent variable

y = Value of the dependent variable

The formula for correlation test is as shown below;

$$t = \frac{r}{\sqrt{\frac{1-r^2}{n-2}}}$$

Calculation

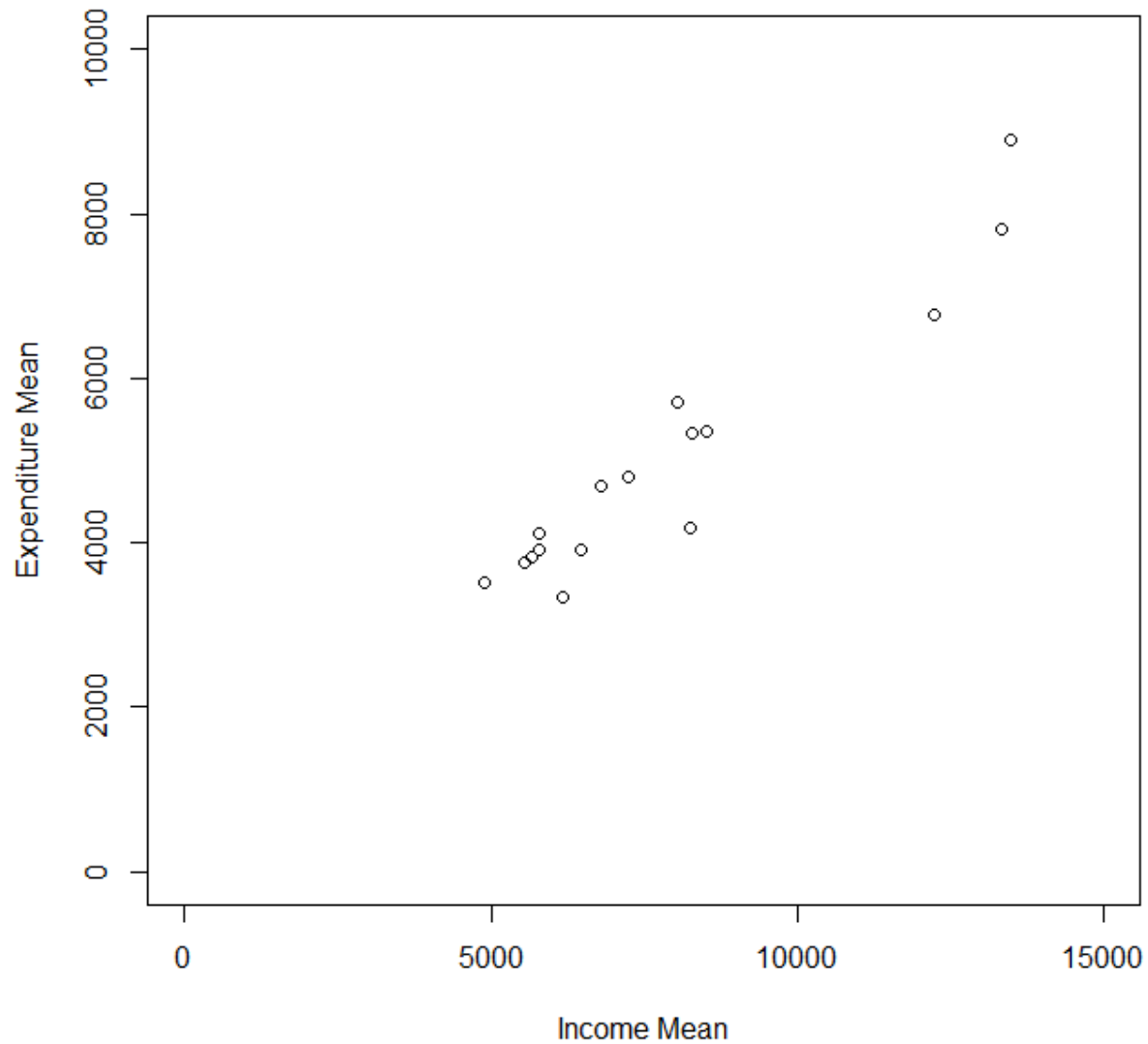


Figure 1 : Scatter Plot for Income Mean and Expenditure Mean

The scatter plot in Figure 1 illustrates the relationship between mean income and mean expenditure across different states. Each point represents a state, with mean income on the x-axis and mean expenditure on the y-axis. The plot shows a positive trend, indicating that as mean income increases, mean expenditure also tends to increase. This suggests a positive linear relationship between the two variables, where higher average incomes are associated with higher average expenditures. However, the strength and significance of this relationship should be confirmed with statistical tests.

Sample Correlation Coefficient

```
> r = cor(x,y)
> r
[1] 0.9562297
```

Figure 2 : The result of r from R Studio

From R Studio, we obtain the value of $r = 0.856$

Hypothesis Statement:

$H_0 : \rho = 0$ (no linear correlation)

$H_1 : \rho \neq 0$ (linear correlation exists)

Test Statistic:

A significant level of 0.05 is used to test the claim that there is linear correlation between income mean and expenditure mean of each state. From the RStudio, we obtain the value of $t = 12.227$, the degree of freedom, $df = 14$ and $p\text{-value} = 7.34932e-09$. From the triola table, $t_{0.025,14} = 2.145$.

```
Degrees of freedom (df): 14
> cat("t-value:", t_value, "\n")
t-value: 12.22719
```

Figure 3 : Degree of Freedom and t value from R Studio

```
> # Print the p-value
> cat("p-value:", p_value, "\n")
p-value: 7.34932e-09
```

Figure 4 : P - value from R Studio

Conclusion

Since $t > t_{0.025,14}$ or $p\text{-value} < \alpha$, reject null hypothesis. There is sufficient evidence to claim that there is a linear relationship between income mean and expenditure mean of each state. This suggests that other factors, such as regional cost of living, state-specific economic conditions, and cultural differences, play more significant roles in determining expenditure patterns.

3.3 Regression Test

In this analysis, we will be using variable expenditure mean and income mean to test whether expenditure means depend on income mean or otherwise. We will be using income mean as independent variable (x) and expenditure mean as dependent variable (y). Since we used the simple linear regression, changes in expenditure mean are assumed by the changes in income mean.

Population linear regression

The population regression model ;

The diagram shows the population linear regression model equation $y = \beta_0 + \beta_1 x + \epsilon$ enclosed in a yellow box. Labels with arrows point to the components of the equation: 'Dependent Variable' points to y ; 'Population y intercept' points to β_0 ; 'Population Slope Coefficient' points to β_1 ; 'Independent Variable' points to x ; and 'Random Error term, or residual' points to ϵ . Below the equation, a bracket under $\beta_0 + \beta_1 x$ is labeled 'Linear component', and a bracket under ϵ is labeled 'Random Error component'.

Linear regression assumptions

- Error values are statistically independent
- Error values are normally distributed for any given value of x
- The probability distribution of the errors is normal
- The probability distribution of the errors has constant variance
- The underlying relationship between the x variable and the y variable is linear

Estimated regression model

The diagram shows the estimated regression model equation $\hat{y}_i = b_0 + b_1 x$ enclosed in a yellow box. Labels with arrows point to the components: 'Estimated (or predicted) y value' points to $\hat{y}_i$; 'Estimate of the regression intercept' points to b_0 ; 'Estimate of the regression slope' points to b_1 ; and 'Independent variable' points to x .

The individual random error terms e_i have a mean of zero.

Test statistic formula

$$t = \frac{b_1 - \beta_1}{s_{b_1}}$$

Where,

b_1 = Sample regression slope coefficient

β_1 = Hypothesized slope

s_{b_1} = Estimator of the standard error of the slope

Coefficient of determination, R^2

The coefficient of determination is the portion of the total variation in the dependent variable that is explained by variation in the independent variable.

$$R^2 = \frac{SSR}{SST} \quad \text{where} \quad 0 \leq R^2 \leq 1$$

$$R^2 = \frac{SSR}{SST} = \frac{\text{sum of squares explained by regression}}{\text{total sum of squares}}$$

Calculation

Figure shows the scatter plot for the relationship between income mean and expenditure mean.

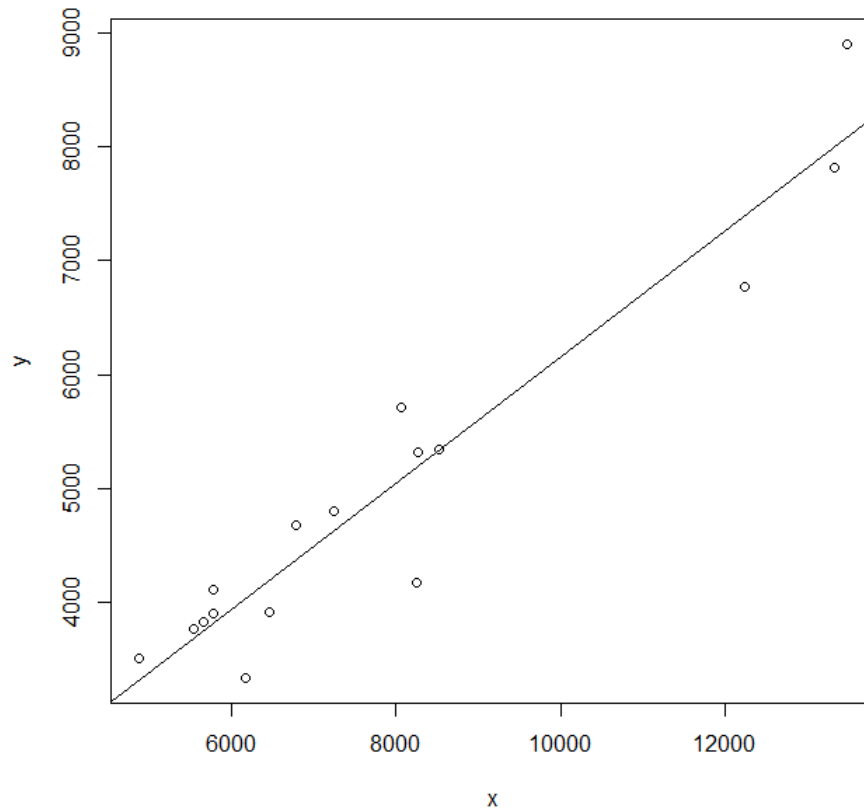


Figure 5 : Scatter Plot

From Figure 5, we can see that income mean and expenditure mean has a positive linear relationship. The best fit line shown with an upward inclining indicates the positiveness of the relationship.

Estimated regression linear model

```
Coefficients:  
(Intercept)      x  
    604.1036    0.5553
```

Figure 6 : Result from R Studio

From the figure, we can conclude that the regression linear model is as follows;

$$\hat{Y} = 604.104 + 0.555x$$

In this model, the intercept (604.104) is the expected expenditure when the income is zero, serving as a baseline expenditure. The slope (0.555) indicates the relationship between income and expenditure; specifically, for every additional unit of income, the expenditure is expected to increase by 0.555 units. This positive slope suggests a direct relationship where higher income leads to higher expenditure.

For the coefficient of determination, R^2 , we obtain the value is 0.914 from R Studio.

```
Residual standard error: 487.6
Multiple R-squared:  0.9144,
F-statistic: 149.5 on 1 and 14
```

Figure 7 : Coefficient of Determination from R Studio

Since $0 < R^2 < 1$, this shows that this high R^2 value suggests a strong positive linear relationship between income and expenditure, meaning that the model fits the data well and income is a significant predictor of expenditure. However, it also implies that 8.6% of the variability in expenditure is due to other factors not accounted for by the model.

Hypothesis Statement

$H_0 : \beta_1 = 0$ (no linear relationship)

$H_1 : \beta_1 \neq 0$ (linear relationship exists)

Test Statistic

A level of significance of 0.05 is used to test the claim that there is no linear relationship between the income mean and expenditure mean. From the R Studio, we obtain the value of sample regression slope coefficient, $b_1 = 0.555$, standard error of the slope, $S_{b1} = 0.045$, test statistic value, $t = 12.227$ and p-value = $7.349e-09$. Degree of freedom is 14 since $n = 16$. From triola table, the critical value, $t_{\alpha/2} = 2.145$.

```
Coefficients:
      Estimate Std. Error t value Pr(>|t|)
(Intercept) 604.10360   379.05369    1.594    0.133
x           0.55533     0.04542   12.227 7.35e-09 ***
```

Figure 8 : Results obtained from R Studio

Conclusion

Since $t > t_{0.025,14}$ or $p\text{-value} < \alpha$, reject null hypothesis. There is sufficient evidence to claim that there is a linear relationship between income mean and expenditure mean. This suggests that other factors, such as regional cost of living, economic conditions, and cultural spending habits, may have a greater influence on expenditure patterns than income alone.

3.4 ANOVA Test

The purpose of the ANOVA test we conducted is to test for significant differences between means of income for four different divisions in Malaysia which are on the East Coast, North Zone of West coast, Middle Zone of West coast and South Zone of West coast . We use a 0.05 significance level to test the null hypothesis that the income for people in different divisions of Malaysia will have the same mean. Due to the same sample size of each population being compared, we apply One-way ANOVA with the Equal Sample Sizes technique. From the data, frequency (n), mean ($\bar{x}$), standard deviation (s) are calculated.

1. Hypothesis statement:

$$H_0 : \mu_1 = \mu_2 = \mu_3 = \mu_4$$

H_1 : at least one mean is different

From the calculation, we know that:

	East Coast	West Coast (North zone)	West Coast (Middle Zone)	West Coat (South Zone)
Frequency	$n_1 = 2$	$n_2 = 2$	$n_3 = 2$	$n_4 = 2$
Mean	$\bar{x}_1 = 5331$	$\bar{x}_2 = 6908.5$	$\bar{x}_3 = 12779$	$\bar{x}_4 = 7422.5$
Standard Deviation	$s_1 = 630.74$	$s_2 = 1921.21$	$s_3 = 772.16$	$s_4 = 897.32$

2. Find variance between samples

By using RStudio, the mean between samples is 8110.25,

<code>overall_mean</code>	8110.25
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standard deviation between samples is 3237.22,

<code>sd_between</code>	3237.21383548672
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and variance between samples is 20959106.83

<code>variance_between</code>	20959106.8333333
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3. Find variance within samples

By using RStudio, the variance within samples is 1372573.76

<code>variance_within</code>	<code>1372573.764925</code>
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4. Given $\alpha = 0.05$. The test statistic, F can be calculated using the formula:

$$F = \frac{\text{variance between sample}}{\text{variance within sample}} = \frac{nS_{\bar{X}}^2}{S_P^2}$$

By using RStudio, the test statistic, F = 15.2699

<code>F_statistic</code>	<code>15.2699311096614</code>
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5. Find numerator and denominator degree of freedom

By using RStudio, the numerator is 3

<code>numerator</code>	<code>3</code>
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And the denominator is 4

<code>denominator</code>	<code>4</code>
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6. Find p-value and critical value.

By using RStudio, the p-value = 0.012

<code>p_value</code>	<code>0.0117631057225902</code>
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Critical value, F-critical value with $\alpha = 0.05$ from F-distribution table is 6.5914

f. alpha	6.59138211642558
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7. Conclusion:

Since $F > F\text{-critical value}$ ($15.3699 > 6.5914$), we reject the null hypothesis. The p-value also shows to reject the null hypothesis since $p\text{-value} = 0.012 < \alpha = 0.05$. There is no sufficient evidence to support the claim that the income for people in different divisions of Malaysia will have the same mean.

Conclusion

In the end, there are significant distinctions between Peninsular Malaysia and East Malaysia as a result of our examination of income and spending data from several Malaysian states. We were able to determine important economic trends as well as variations in wealth and cost of living by concentrating on mean income and mean expenses. Higher rates of poverty in some places draw attention to the need for specific measures to address these problems, which is important for decision-makers in economic planning and resource allocation as well as policymakers.

We found significant links and patterns in the data through our statistical analysis, which included hypothesis testing, correlation, and ANOVA. These understandings serve as a basis for creating policies that encourage equal regional development and lessen economic inequality. All things considered, our research highlights the significance of ongoing data gathering and analysis for tracking economic trends and assessing the performance of policies, guaranteeing equal development throughout Malaysia's regions.

Appendix

1. Raw Data

date	state	income_mean	income_median	expenditure_mean	gini	poverty
1/1/2022	Johor	8517	6879	5342	0.36646	4.6
1/1/2022	Kedah	5550	4402	3765	0.35938	9
1/1/2022	Kelantan	4885	3614	3505	0.3854	13.2
1/1/2022	Melaka	8057	6210	5707	0.36963	4.2
1/1/2022	Negeri Sembilan	6788	5226	4678	0.36853	4.4
1/1/2022	Pahang	5777	4753	4107	0.3077	6.3
1/1/2022	Pulau Pinang	8267	6502	5322	0.37058	2
1/1/2022	Perak	5779	4494	3903	0.36769	7.5
1/1/2022	Perlis	5664	4713	3834	0.33589	4
1/1/2022	Selangor	12233	9983	6770	0.36123	1.5
1/1/2022	Terengganu	7248	5878	4796	0.32631	6.2
1/1/2022	Sabah	6171	4577	3342	0.39491	19.7
1/1/2022	Sarawak	6457	4978	3915	0.3818	10.8
1/1/2022	W.P. Kuala Lumpur	13325	10234	7823	0.3796	1.4
1/1/2022	W.P. Labuan	8250	6904	4176	0.30028	2.5
1/1/2022	W.P. Putrajaya	13473	10056	8897	0.3678	0.1

2. Updated Data from Raw Data

	A	B	C	D	E	F	G	H
1	date	state	income_mean	income_median	expenditure_mean	gini	poverty	region (Malaysia)
2	1/1/2022	Johor	8517	6879	5342	0.36646	4.6	Peninsular (West Coast)
3	1/1/2022	Kedah	5550	4402	3765	0.35938	9	Peninsular (West Coast)
4	1/1/2022	Kelantan	4885	3614	3505	0.3854	13.2	Peninsular (East Coast)
5	1/1/2022	Melaka	8057	6210	5707	0.36963	4.2	Peninsular (West Coast)
6	1/1/2022	Negeri Sembilan	6788	5226	4678	0.36853	4.4	Peninsular (West Coast)
7	1/1/2022	Pahang	5777	4753	4107	0.3077	6.3	Peninsular (East Coast)
8	1/1/2022	Pulau Pinang	8267	6502	5322	0.37058	2	Peninsular (West Coast)
9	1/1/2022	Perak	5779	4494	3903	0.36769	7.5	Peninsular (West Coast)
10	1/1/2022	Perlis	5664	4713	3834	0.33589	4	Peninsular (West Coast)
11	1/1/2022	Selangor	12233	9983	6770	0.36123	1.5	Peninsular (West Coast)
12	1/1/2022	Terengganu	7248	5878	4796	0.32631	6.2	Peninsular (East Coast)
13	1/1/2022	Sabah	6171	4577	3342	0.39491	19.7	East
14	1/1/2022	Sarawak	6457	4978	3915	0.3818	10.8	East
15	1/1/2022	W.P. Kuala Lumpur	13325	10234	7823	0.3796	1.4	Peninsular (West Coast)
16	1/1/2022	W.P. Labuan	8250	6904	4176	0.30028	2.5	East Malaysia
17	1/1/2022	W.P. Putrajaya	13473	10056	8897	0.3678	0.1	Peninsular (West Coast)
18								

3. LinkedIn -

https://www.linkedin.com/posts/farra-nurzahin-zaharil-anuar-44107329a_seci1143-02-pr

obability-and-statistics-data-activity-7214295734567723009-o3zk?utm_source=share&utm_medium=member_ios